



After opening the black box: Meta-dehumanization matters in algorithm recommendation aversion

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ABSTRACT

Perceptions of algorithms as opaque, commonly referred to as the black box problem, can make people reluctant to accept a recommendation from an algorithm rather than a human. Interventions that enhance people's subjective understanding of algorithms have been shown to reduce this aversion. However, across four preregistered studies ($N = 960$), we found that in the online shopping context, after explaining the algorithm recommendation process (versus human recommendation), users felt dehumanized and thus averse to algorithms (Study 1). This effect persisted, regardless of the type of algorithm (i.e., conventional algorithms or large language models; Study 2) or recommended product (i.e., search or experience products; Study 3). Notably, considering large language models (versus conventional algorithms) as the recommendation agent (Study 2) and framing algorithm recommendation as consumer-serving (versus website-serving; Study 4) mitigated algorithm aversion caused by meta-dehumanization. Our findings contribute to ongoing discussions on algorithm transparency, enrich the literature on human–algorithm interaction, and provide practical insights for encouraging algorithm adoption.

1. Introduction

Artificial intelligence (AI) has become ubiquitous in our daily life, such as algorithmic recommendations in online shopping (Longoni & Cian, 2022). However, as certain algorithms, such as deep learning models, have become increasingly accurate in their predictions, they have also become opaquer, exacerbating the challenges users face in understanding how algorithms generate recommendations (i.e., the “black box” problem; Rai, 2020; Von Eschenbach, 2021). A lack of transparency in algorithms can potentially deter users from embracing these technologies fully (Cadario et al., 2021; De Freitas et al., 2023; Yeomans et al., 2019), which underscores the need to open the “black box” of algorithms through advocating for greater transparency in algorithmic processes (Clarke, 2019; Goodman & Flaxman, 2017; Watson & Nations, 2019).

Previous studies have shown that explaining how algorithmic recommendations are generated can help reduce algorithm aversion (Cadario et al., 2021; Yeomans et al., 2019). Algorithms primarily rely on calculations as the fundamental mechanism for their functioning; therefore, an intriguing question arises: is it beneficial to open the black

box of algorithms by explaining this calculative nature to users? We posit that when users understand that algorithms base their recommendations on calculations, they may feel dehumanized and consequently exhibit aversion toward algorithmic recommendations. Thus, the current research aims to empirically investigate this phenomenon, with the aim of enhancing the theoretical understanding of algorithm aversion and offering practical insights for promoting transparent algorithms.

1.1. Algorithms as black boxes and algorithm aversion

The rise of deep learning models has made algorithms significantly more effective but concurrently rendered them opaquer, leading to diminished trust among users (Von Eschenbach, 2021). For example, large language models (LLMs) offer personalized product recommendations in online shopping through interactive conversations with users; however, users often cannot comprehend the rationale behind these recommendations. More importantly, users are not necessarily more, and may even be less, willing to adopt recommendations from algorithms than from human experts. This phenomenon, called algorithm

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aversion (Dietvorst et al., 2015; Jussupow et al., 2020), has been widely observed across multiple contexts, including organizational settings, healthcare, and electronic commerce (e-commerce; Dietvorst et al., 2015; Highhouse, 2008; Longoni et al., 2019; Longoni & Cian, 2022; Luo et al., 2019; Önköl et al., 2009; Prah & Van Swol, 2017; Reich et al., 2023; Xie et al., 2022; Yeomans et al., 2019).

Previous research has explored the psychological factors underlying algorithm aversion (for a recent review, see De Freitas et al., 2023). For example, people exhibit lower tolerance for errors made by algorithms compared to those made by humans, making them hesitant to adopt algorithms (Dietvorst et al., 2015). Moreover, algorithms are often perceived as lacking the ability to learn from their errors, rendering them less appealing than their human counterparts (Reich et al., 2023). Concerns that algorithms may neglect individuals' uniqueness can lead people to prefer services delivered by humans rather than algorithms (Longoni et al., 2019; Purcell & Bonnefon, 2023). Despite algorithms demonstrating superior performance to that of humans in various domains (Castelo et al., 2019; Meehl, 1954; Yeomans et al., 2019), they can generate and spread misinformation in some instances. Importantly, the perceived lack of transparency in algorithms fuels concerns regarding the credibility and accuracy of the information they generate (Shin, 2022), consequently undermining users' trust in them (Shin et al., 2022).

Various strategies have been proposed to enhance algorithm transparency to address algorithm aversion, including the development of explainable AI (Adadi & Berrada, 2018; Rai, 2020). Explainable AI addresses the need for algorithms that can explain their decisions in a way humans will understand (Rai, 2020). Consistent with this approach, providing explanations on how algorithmic decisions are made has been found to increase users' understandings of algorithms and thus reduce algorithm aversion. For example, interventions clarifying that medical algorithms diagnose skin cancer by comparing a mole to a library of benign and cancerous moles have been successful in improving people's understanding of the diagnosis process and their willingness to use algorithmic healthcare providers (Cadario et al., 2021). Similarly, explaining how algorithms recommend jokes has been shown to reduce aversion by increasing users' subjective understanding of the recommendation process (Yeomans et al., 2019).

However, while these interventions are helpful to some extent, they ignore the different inner workings of algorithmic and human decision-making. In a previous study, Cadario et al. (2021) used the same visuals to illustrate how medical algorithms and doctors provide healthcare service, with the only difference being that the provider was labeled either as "doctor" or "algorithm." Unlike human recommendations, which are mainly based on the recommenders' knowledge, expertise, or experience, algorithm recommendations involve calculations. This leads to the question of whether further opening the "black box" of algorithms by introducing their workings compared to human recommendations is beneficial. We posit that, compared to expertise-based human recommendations, calculation-based algorithm recommendations make people feel more dehumanized and thus less willing to adopt them. Accordingly, meta-dehumanization would account for the differences in attitudes toward algorithm versus human recommendations.

1.2. Meta-dehumanization drives algorithm aversion

Meta-dehumanization refers to perceiving oneself as being treated as less than human (Kteily et al., 2016). Echoing the twofold structure of dehumanization (i.e., likening others to animals or machines; Haslam & Loughnan, 2014), meta-dehumanization takes two forms (Demoulin et al., 2021). Animalistic meta-dehumanization entails perceiving that one's uniqueness as a human being, separate from animals (e.g., civility and rationality), is downplayed, whereas mechanistic meta-dehumanization involves perceiving that human nature distinguishing people from machines (e.g., emotionality and cognitive openness) is downplayed. Previous research has indicated that

meta-dehumanization occurs in social contexts where calculative thinking (i.e., "making decisions by computing the costs and benefits"; Belmi & Schroeder, 2021, p. 384) is salient (Fiske, 1992). For example, employees perceiving that their supervisors consider them a number (e.g., the profit they make) is a hallmark of meta-dehumanization in organizational settings (Bell & Khoury, 2011; Belmi & Schroeder, 2021; Brison et al., 2022; Caesens et al., 2017).

We propose that algorithm recommendations may elicit meta-dehumanization in users when the calculative nature of algorithms is introduced. Anecdotal evidence suggests that using algorithms to evaluate people may be dehumanizing (Dawes, 1979; Grove & Meehl, 1996; Lee, 2018). To make recommendations, algorithms use calculations to dig out consumer needs and preferences from their profiles (e.g., historical choices and social networks). Given that needs and preferences are integral aspects of humanity (Deci & Ryan, 2000; Lopez et al., 2021; Schroeder & Epley, 2020), individuals may feel dehumanized if their psychological needs are disregarded or subjective preferences are overlooked. Thus, the calculative nature of algorithm recommendations may lead consumers to feel dehumanized in a way that their humanity is calculated and thus devalued (Christoff, 2014; Demoulin et al., 2021; Fontesse et al., 2019; Lopez et al., 2021; Schroeder & Epley, 2020).

Meta-dehumanization promotes aversion. People often develop distant perceptions of (Bastian & Haslam, 2010; Kteily et al., 2016) and express hostility toward (Kteily & Bruneau, 2017) perpetrators who dehumanize them. Similarly, they tend to adopt avoidance-oriented coping strategies (e.g., denial) in contexts where they feel dehumanized (Demoulin et al., 2021). Notably, the perpetrators need not to be human agents but can encompass abstract non-human entities. For example, feeling dehumanized in organizational contexts prompts employees to distance themselves from organizations, thereby increasing turnover intentions (Bell & Khoury, 2016; Caesens et al., 2019). Therefore, when users feel dehumanized by algorithms, they may be reluctant to accept their recommendations.

1.3. The present research

We hypothesized that, compared to expertise-based human recommendations, calculation-based algorithm recommendations would elicit greater meta-dehumanization and thus less willingness to adopt among users. Considering the potential benefits of algorithm recommendations in e-commerce (Komiak & Benbasat, 2006; Xie et al., 2022), whether algorithm aversion persists even after algorithmic processes have been explained must be further investigated. Therefore, we tested our hypothesis within e-commerce contexts.

Our general approach across four studies involved describing calculation-based algorithm recommendations and expertise-based human recommendations in e-commerce contexts and assessing participants' meta-dehumanization and algorithm aversion, operationalized as less intention to use algorithms (versus humans) for recommendations. Study 1 examined how recommendation agents influence meta-dehumanization and people's subsequent intention to use these agents. Study 2 extended Study 1 by distinguishing between conventional and more interactive algorithms (i.e., LLMs), exploring whether conventional algorithms elicit different levels of meta-dehumanization and subsequent use intention compared to LLMs. To examine whether the observed effects in Studies 1 and 2 depend on the recommended product type, Study 3 tested our hypothesis across different types (search or experience) of recommended products. Considering the important role of meta-dehumanization in algorithm aversion, interventions that aim to reduce meta-dehumanization are expected to alleviate algorithm aversion. Therefore, Study 4 examined whether framing algorithms as consumer-serving (versus website-serving), as a means to highlight social connections between algorithms and consumers, can provide a buffer against algorithm aversion.

We reported all measures, conditions, and data exclusions in the studies. We preregistered the hypotheses, design, and data analysis plans

of all four studies (Study 1: https://osf.io/cxn6p/?view_only=b311b85b70004652b39eedaffc25ceca; Study 2: https://osf.io/8guq7/?view_only=badb9fbf0d7946d1af6e5954ace1f86a; Study 3: https://osf.io/xugqt/?view_only=672a6c0a4ab843daadd29c7f6bd394cc; Study 4: https://osf.io/8ca72/?view_only=e13f448ac87b409e9813ce67a667c4de). We made all data available at https://osf.io/2bfwx/?view_only=d992420c230448a186ada66a39547795. In the Supplementary Material, we reported the stimulus materials, a pilot study for Study 3, a supplemental study and supplemental analyses.

2. Study 1

Study 1 aimed to provide primary evidence for our hypothesis. We presented participants with randomly distributed explanations of how algorithms or experts make recommendations in an e-commerce context. We predicted that algorithm (versus expert) recommendations would elicit more meta-dehumanization and thus less intention to use.

2.1. Method

2.1.1. Participants

Using the smallest effect size of interest (Lakens, 2014; $d = 0.40$) and $\alpha = .05$ (two-tailed), a sample size of 200 would be sufficient to achieve 80% power to detect the effect of recommendation agents on meta-dehumanization in a two-condition between-subject experiment. We conservatively recruited 248 Chinese participants (151 women and 97 men, $M_{age} = 29.69$ years, $SD_{age} = 8.44$) via Credamo (<https://credamo.com>) and paid 2 CNY (0.29 USD) to each participant. No participants were excluded following the preregistered criteria (i.e., being an outlier or failing the attention check items). Participants were randomly assigned to either the algorithm ($n = 125$) or expert ($n = 123$) recommendation condition.

2.1.2. Procedure and materials

2.1.2.1. Recommendation agent manipulation. After providing informed consent, participants read an introduction to the functions and mechanisms of recommendation systems in the online shopping context. In the *algorithm recommendation* condition, participants read that the algorithm provides recommendations by quantifying and calculating consumers' preference profiles using mathematical models. In the *expert recommendation* condition, participants were informed that experts use experience and knowledge to provide recommendations by learning and understanding consumers' preferences. As both a manipulation and an attention check, after reading the material, participants were asked to select the recommendation agent referred to in the introduction.

2.1.2.2. Meta-dehumanization. Next, participants completed an adapted 10-item meta-dehumanization scale from Demoulin et al. (2021). The scale comprises two subscales, animalistic and mechanistic meta-dehumanization, with mixed items presented in random order. Five items measure animalistic meta-dehumanization (e.g., "I think that algorithm/expert recommendation systems devalue my intelligence"), and five items measure mechanistic meta-dehumanization (e.g., "I think that algorithm/expert recommendation systems devalue my subjectivity"). All items were rated on a seven-point scale (1 = *completely disagree*, 7 = *completely agree*). We averaged ratings to create animalistic ($\alpha = .90$) and mechanistic ($\alpha = .90$) meta-dehumanization indices. Higher ratings indicated higher levels of meta-dehumanization.

2.1.2.3. Intention to use. Finally, participants completed an adapted three-item behavioral intention to use scale developed by Moon and Kim (2001; e.g., "I will use algorithm/expert recommendation systems on a regular basis in the future"; 1 = *completely disagree*, 7 = *completely agree*), and an item assessing the extent to which they were willing to adopt the

recommendation system, rated on a seven-point scale (1 = *completely unwilling*, 7 = *completely willing*). We averaged the ratings for the four items ($\alpha = .90$) to create the intention to use index. Higher ratings indicated higher levels of intention to use.

2.2. Results and discussion

2.2.1. Effects of recommendation agent

Participants in the algorithm recommendation condition ($M = 3.75$, $SD = 1.52$) reported greater mechanistic meta-dehumanization than those in the expert recommendation condition ($M = 3.07$, $SD = 1.37$), $t(246) = 3.70$, $p < .001$, $d = 0.47$, 95%CI [0.22, 0.72]. However, participants across both conditions did not differ in animalistic meta-dehumanization ($M = 2.96$, $SD = 1.28$ vs. $M = 2.65$, $SD = 1.26$), $t(246) = 1.92$, $p = .057$, $d = 0.24$, 95%CI [-0.01, 0.49], or intention to use ($M = 4.68$, $SD = 1.28$ vs. $M = 4.65$, $SD = 1.26$), $t(246) = 0.18$, $p = .855$, $d = 0.02$, 95%CI [-0.23, 0.27]. As significant indirect effects can occur without significant total effects (Hayes, 2009; Rucker et al., 2011), we proceeded to test the mediating role of meta-dehumanization in the effect of recommendation agents on intention to use.

2.2.2. Mediating role of meta-dehumanization

Using Mplus 7.4, we tested a parallel mediation model with mechanistic and animalistic meta-dehumanization as the mediators. As shown in Fig. 1, algorithm (versus expert) recommendations increased mechanistic meta-dehumanization ($B = 0.68$, 95%CI [0.31, 1.03], $SE = 0.18$, $t = 3.74$, $p < .001$), which, in turn, predicted less intention to use ($B = -0.56$, 95%CI [-0.70, -0.42], $SE = 0.07$, $t = -7.92$, $p < .001$). Algorithm (versus expert) recommendations did not increase animalistic meta-dehumanization ($B = 0.31$, 95%CI [-0.01, 0.62], $SE = 0.16$, $t = 1.93$, $p = .055$), and animalistic meta-dehumanization did not predict intention to use ($B = -0.03$, 95%CI [-0.21, 0.14], $SE = 0.09$, $t = -0.30$, $p = .768$). The indirect pathway via mechanistic meta-dehumanization was significant ($ab = -0.38$, bootstrap 95%CI [-0.63, -0.18], $SE = 0.11$); however, the indirect effect via animalistic meta-dehumanization was not ($ab = -0.01$, bootstrap 95%CI [-0.09, 0.04], $SE = 0.03$).

Furthermore, the total effect of recommendation agent on intention to use was non-significant ($B = 0.03$, 95%CI [-0.27, 0.35], $SE = 0.16$, $t = 0.18$, $p = .854$). Notably, when controlling for animalistic and mechanistic meta-dehumanization, algorithm (versus expert) recommendations elicited greater intention to use, $B = 0.42$, 95%CI [0.18, 0.66], $SE = 0.12$, $t = 3.50$, $p < .001$. Accordingly, controlling for meta-dehumanization made the direct effect of recommendation agent more positive than the total effect.

Study 1 provided preliminary support for the hypothesis. When opening the black box by explaining recommendation processes,

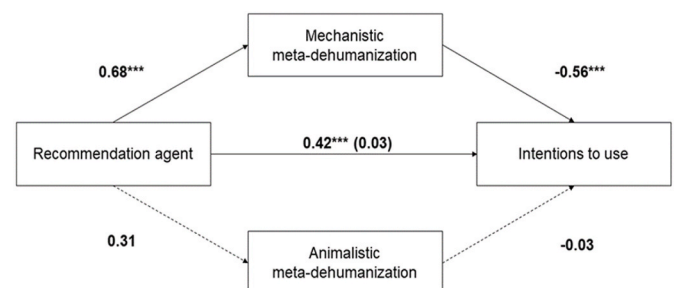


Fig. 1. Effect of recommendation agent on intention to use via meta-dehumanization in Study 1.

Note. Recommendation agent manipulation coding: algorithm recommendation condition = 1, expert recommendation condition = 0. Coefficients are unstandardized. The coefficient in parentheses is the total effects of recommendation agent on intention to use. To enhance figure clarity, we omitted the correlation between animalistic and mechanistic meta-dehumanization ($B = 1.43$, 95%CI [1.18, 1.73], $SE = 0.14$, $p < .001$). *** $p < .001$.

algorithms (versus experts) provided recommendations that increased mechanistic meta-dehumanization and thus reduced consumers' use intention. The results echo prior discussions of the association between algorithm aversion and perceptions of algorithms not treating people with interpersonal warmth (Grove & Meehl, 1996), a hallmark of mechanistic dehumanization (Haslam, 2006). Notably, animalistic meta-dehumanization could not account for aversion to algorithm recommendation. This might be because animalistic form of meta-dehumanization is more likely to occur when perceiving oneself as being likened to animals rather than inanimate substances (e.g., numbers or machines; Demoulin et al., 2021). Accordingly, we only included mechanistic meta-dehumanization in the models in subsequent studies.

Interestingly, we did not find a significant total effect of recommendation agent on intention to use algorithm recommendation. This non-significant total effect ruled out the alternative pathway of algorithm (versus expert) recommendation $\rightarrow$ intention to use $\rightarrow$ meta-dehumanization, indicating that meta-dehumanization acts as a predictor, rather than an outcome, of algorithm aversion. Importantly, this total effect is composed of a negative indirect effect of recommendation agent on intention to use via increased meta-dehumanization and a positive direct. Considering the negative indirect effect is directionally opposite to the positive direct effect, this pattern accounts to a suppression situation (Paulhus et al., 2004; Rucker et al., 2011), and in the presented suppression framework, mechanistic meta-dehumanization suppressed the positive direct effect of algorithm (versus expert) recommendations on intention to use, indicating that mechanistic meta-dehumanization is a key mechanism underlying algorithm aversion.

3. Study 2

Study 2 aimed to extend Study 1 by exploring whether the mediating role of mechanistic meta-dehumanization depends on the algorithm type. Systems powered by conventional recommendation algorithms rely on consumer data (e.g., purchase history) and are criticized as having poor interactivity (Gao et al., 2023). Recently, novel recommendation systems powered by LLMs have been broadly deployed. LLMs are deep learning algorithms that can process human language and interact with consumers naturally based on pre-trained textual data (Fan et al., 2023). LLM recommendations work by transforming human language (e.g., consumers' needs or preferences) into textual data and generating content (e.g., recommendations) after computing the data. LLMs provide recommendations to consumers in a more conversational manner compared with conventional algorithms. Given that interactivity and language are important social cues (Ossadnik et al., 2023), recommendations provided by LLMs are expected to involve more social engagement than those provided by conventional algorithms. Therefore, people may experience less mechanistic meta-dehumanization when receiving recommendations from LLMs rather than from conventional algorithms. Therefore, we predicted that the mediating effect of mechanistic meta-dehumanization would be weaker in aversion to LLMs compared with aversion to conventional algorithms.

3.1. Method

3.1.1. Participants

Using the effect size ($d = 0.47$) obtained in Study 1 and $\alpha = .05$ (two-tailed), a sample size of 146 (73 per condition) was determined to be sufficient to achieve 80% power to detect the effect of recommendation agent (conventional algorithm versus expert) on meta-dehumanization in a two-condition experiment. However, because we included one more condition (i.e., LLM condition), 73 additional participants were

required. We conservatively recruited 300 Chinese participants via Credamo and paid 2 CNY (0.29 USD) to each participant. We excluded two participants following the preregistered criteria (i.e., being an outlier or failing the attention check items),¹ and the final sample included 298 participants (176 women and 122 men, $M_{age} = 29.78$ years, $SD_{age} = 9.41$). Participants were randomly assigned to one of three conditions: conventional algorithm ($n = 100$), LLM ($n = 100$), or expert ($n = 98$) recommendation condition.

3.1.2. Procedure and materials

3.1.2.1. Recommendation agent manipulation. After providing informed consent, all participants read descriptions of how recommendation systems are used in online shopping contexts before completing two attention check questions (e.g., "To what domain are these recommendation agents applied?"). To clarify the variability of the recommendation agent, we described three agents on the same page.

Next, participants were asked to rate only one of the three recommendation agents, depending on the condition to which they were assigned. To ensure that the participants understand how the recommendation agent they rated worked, they read the same description of the specific recommendation agent as before. Specifically, in the *conventional algorithm recommendation* condition, participants read that a conventional recommendation system collects consumers' personal information and purchase history and transforms that data into various parameters. These parameters are used to analyze consumers' preferences based on mathematical models, and the system calculates the similarity between consumers' interests and product information to provide recommendations. In the *LLM recommendation* condition, participants read that an LLM recommendation system allows consumers to express their needs or requests to the system via chat windows. The system decodes consumers natural language based on statistical models constructed by pre-learned textual data, and predicts consumers' preferences to make recommendations. In the *expert recommendation* condition, participants read that an expert recommendation system works based on real-time conversations between customer service staff and consumers. Customer service staff can understand consumers' needs based on their experience and knowledge and predict consumers' interests and preferences to offer recommendations. To simulate the work of recommendation agents in the real world, participants also read an online shopping scenario in which a consumer was planning to buy gifts for Teacher's Day, and viewed a figure depicting how the consumer received recommendations from the agent (see the Supplementary Material). After reading the introduction and scenarios, participants completed three attention check questions (e.g., "What type of recommendation agent did we ask you to rate?").

3.1.2.2. Meta-dehumanization. Participants completed the same measure of mechanistic meta-dehumanization ($\alpha = .95$) as in Study 1.

3.1.2.3. Intention to use. Participants completed the same measure of intention to use ($\alpha = .94$) as in Study 1.

3.2. Results and discussion

3.2.1. Effects of recommendation agent

We conducted ANOVAs on mechanistic meta-dehumanization and intention to use. Mechanistic meta-dehumanization significantly varied across conditions, $F(2, 194.53) = 8.19, p < .001, \eta^2 = 0.06, 95\%CI [0.02, 0.10]$. Planned comparisons revealed that mechanistic meta-dehumanization scores in the conventional algorithm recommendation

¹ Across Studies 2–4, the key results remained unchanged regardless of whether outliers or participants who failed attention checks were excluded.

condition ($M = 3.70$, $SD = 1.83$) were higher than those in both the LLM ($M = 3.18$, $SD = 1.44$), $t(187.72) = 2.23$, $p = .027$, $d = 0.32$, 95%CI [0.04, 0.59], and the expert recommendation condition ($M = 2.77$, $SD = 1.37$), $t(183.75) = 4.04$, $p < .001$, $d = 0.57$, 95%CI [0.29, 0.85]. Mechanistic meta-dehumanization scores were higher in the LLM recommendation condition than in the expert recommendation condition, $t(196) = 2.03$, $p = .043$, $d = 0.29$, 95%CI [0.01, 0.57].

Moreover, intention to use significantly varied across conditions, $F(2, 196.07) = 6.50$, $p = .002$, $\eta^2 = 0.04$, 95%CI [0.01, 0.08]. Planned comparisons revealed that intention to use scores in the conventional algorithm recommendation condition ($M = 4.61$, $SD = 1.62$) were lower than those in the expert recommendation condition ($M = 5.28$, $SD = 1.37$), $t(192.02) = -3.12$, $p = .002$, $d = -0.44$, 95%CI [-0.72, -0.16], but similar to those in the LLM recommendation condition ($M = 4.66$, $SD = 1.51$), $t(198) = -0.21$, $p = .831$, $d = -0.03$, 95%CI [-0.31, 0.25]. In addition, intention to use scores in the LLM recommendation condition were lower than those in the expert recommendation condition, $t(196) = -3.01$, $p = .003$, $d = -0.43$, 95%CI [-0.72, -0.15].

3.2.2. Mediating role of mechanistic meta-dehumanization

We tested the mediating role of mechanistic meta-dehumanization using Mplus 7.4. We created two dummy codes for the three recommendation agent conditions: X_1 (conventional algorithm = 1, LLM = 0, expert = 0) and X_2 (conventional algorithm = 0, LLM = 1, expert = 0), with the expert recommendation condition as a reference category. As Fig. 2 shows, both X_1 ($B = 0.93$, 95%CI [0.48, 1.37], $SE = 0.23$, $t = 4.05$, $p < .001$) and X_2 ($B = 0.41$, 95%CI [0.01, 0.79], $SE = 0.20$, $t = 2.04$, $p = .042$) increased mechanistic meta-dehumanization. Mechanistic meta-dehumanization predicted less intention to use ($B = -0.56$, 95%CI [-0.69, -0.44], $SE = 0.06$, $t = -8.89$, $p < .001$). The indirect effects of both X_1 ($ab = -0.52$, bootstrap 95%CI [-0.83, -0.27], $SE = 0.14$) and X_2 ($ab = -0.23$, bootstrap 95%CI [-0.48, -0.01], $SE = 0.12$) on intention to use via mechanistic meta-dehumanization were significant. Moreover, the indirect effect of X_1 on intention to use was stronger than that of X_2 , $B = -0.29$, 95%CI [-0.57, -0.09], $SE = 0.13$, $t = -2.22$, $p = .027$.

After controlling for mechanistic meta-dehumanization, the direct effect of X_1 on intention to use was not significant ($B = -0.14$, 95%CI [-0.52, 0.22], $SE = 0.19$, $t = -0.76$, $p = .448$), whereas the total effect was ($B = -0.67$, 95%CI [-1.07, -0.24], $SE = 0.22$, $t = -3.08$, $p = .002$). In contrast, both the direct ($B = -0.39$, 95%CI [-0.73, -0.06], $SE = 0.17$, $t = -2.28$, $p = .022$) and total ($B = -0.62$, 95%CI [-1.02, -0.22], $SE = 0.21$, $t = -3.02$, $p = .003$) effects of X_2 on intention to use were

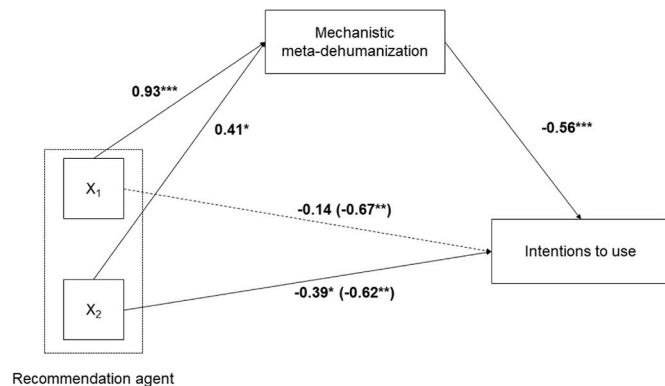


Fig. 2. Effects of recommendation agent on intention to use via mechanistic meta-dehumanization in Study 2.

Note. Recommendation agent manipulation was dummy coded (X_1 : conventional algorithm = 1, LLM = 0, expert = 0; X_2 : conventional algorithm = 0, LLM = 1, expert = 0). Coefficients are unstandardized. Coefficients in parentheses are the total effects of recommendation agent on intention to use. * $p < .05$. ** $p < .01$. *** $p < .001$.

significant.

Study 2 demonstrated that the hypothesis held when the recommendation agent was an LLM. However, because LLMs might make recommendations in a more interactive manner than conventional algorithms do, the mediating role of mechanistic meta-dehumanization was weakened. The results contribute to the generalizability of the hypothesis; regardless of how advanced algorithms are, mechanistic meta-dehumanization may essentially drive algorithm aversion if consumers believe that algorithms treat their needs as data. However, the results also indicated that increasing interactive elements and using human language when providing recommendations might help reduce the meta-dehumanization engendered by algorithms, thereby buffering algorithm aversion.

Unlike Study 1, Study 2 revealed significant total effects of recommendation agent on intention to use recommendation systems. This might be because we specified a context of buying gifts for Teachers' Day. Giving gifts is a key practice of establishing or strengthening social ties (Mead et al., 2011); therefore, the process of selecting gifts should involve interpersonal warmth. If participants perceived algorithms (including conventional and LLM recommendation systems) as dehumanizing or lacking interpersonal warmth (Luo et al., 2019), they may have been less willing to adopt the recommendations that those algorithms provided (versus those offered by customer service staff). We return to this point in the General Discussion.

4. Study 3

Study 3 extended previous studies in two ways. Theoretically, Study 3 specified whether the observed effect depends on the recommended product type, arguably influencing algorithm aversion (Xie et al., 2022). Products can mainly be differentiated into two types: search and experience products (Huang et al., 2009; Nelson, 1970). The determinant attributes of search products (e.g., USB flash drive) are related to objective parameters and can be evaluated before purchase, whereas the characteristics of experience products (e.g., juice) are related to subjective experiences and cannot be observed until after use. Previous research has suggested that people feel algorithm aversion when receiving recommendations experience products because algorithms lack the ability of feeling emotions (Xie et al., 2022). Notably, although some advanced algorithms (e.g., LLMs) appeal to understand humans' feelings, their functions are qualitatively distinct from humans' ones (Wang et al., 2023). However, if algorithms' calculative working processes were a key factor in consumers' feelings of dehumanization, then consumers would resist using any product an algorithm recommended. Therefore, we predicted that the hypothesis would hold regardless of the type of product recommended. Methodologically, Study 3 used a within-subject design that enabled participants to evaluate two recommendation agents simultaneously, which was expected to avoid potential ceiling or floor effects (Longoni et al., 2019).

4.1. Method

4.1.1. Participants

Study 3 used a 2 (recommendation agent: algorithm, expert) $\times$ 2 (product type: search, experience) mixed-factorial design, with recommendation agent as the within-subject factor and product type as the between-subject factor. Using the effect size ($d = 0.47$) obtained in Study 1 and $\alpha = .05$ (two-tailed), a sample size of 38 would be sufficient to achieve 80% power to detect the effect of recommendation agent on meta-dehumanization in a two-condition within-subject experiment. Although we did not expect this moderation pattern, to explore whether the effect of recommendation agent on meta-dehumanization would vary by product type, we recruited 200 Chinese participants via Credamo and paid 2 CNY (0.29 USD) to each participant. We excluded one participant following the preregistered criteria (i.e., being an outlier or failing the attention check items), and the final sample included 199

participants (121 women and 78 men, $M_{age} = 28.92$, $SD_{age} = 8.05$). Participants were randomly assigned to either the search ($n = 99$) or experience ($n = 100$) product condition.

4.1.2. Procedure and materials

4.1.2.1. Recommendation agent manipulation. After providing informed consent, all participants read the introductions on algorithm and expert recommendations as used in Study 1. To clarify the variability of the recommendation agent, we described both recommendation systems on the same page. As a manipulation check, participants indicated the extent to which the algorithm/expert recommendation systems worked based on calculations or experience according to the material (1 = *mainly based on calculations*, 7 = *mainly based on experience*). Higher ratings indicated that the recommendation systems worked more based on experience than calculations.

4.1.2.2. Product type manipulation. Next, participants were randomly assigned to one of two conditions and asked to imagine that they planned to buy a USB flash drive (search product condition) or a dozen bottles of juice (experience product condition). In a pilot study, we selected these two products as typical search and experience products, respectively, and participants were equally familiar with both (see the Supplementary Material). The two products were provided with the same price estimation (i.e., 20–50 CNY) to control for the potential influence of price. Next, participants read that as soon as they went to the shopping website, the algorithm and expert recommendation systems would recommend products. As a manipulation check, participants indicated the extent to which the determinant attributes of the product they would buy were related to objective parameters or subjective experience (1 = *mainly objective parameters*, 7 = *mainly subjective experience*).

4.1.2.3. Meta-dehumanization. After that, participants indicated the extent to which they felt dehumanized by algorithm ($\alpha = .92$) and expert ($\alpha = .91$) recommendations with the same five-item measure of mechanistic meta-dehumanization as in previous studies, respectively. The presentation order of the two scales was randomized.

4.1.2.4. Intention to use. Finally, participants reported their intention to use each recommendation agent based on the same four items as in previous studies. We averaged the relevant items to create respective indices for intention to use algorithm ($\alpha = .92$) and expert ($\alpha = .92$) recommendations.

4.2. Results and discussion

4.2.1. Manipulation checks

Participants believed that algorithm recommendations ($M = 1.20$, $SD = 0.57$) worked less based on experience compared with expert recommendations ($M = 6.74$, $SD = 0.51$), $t(198) = -79.62$, $p < .001$, $d = -5.64$, 95%CI $[-5.78, -5.50]$. In addition, the search product ($M = 2.25$, $SD = 1.70$) was rated as being less related to subjective experience than the experience product ($M = 4.58$, $SD = 2.05$), $t(191.28) = -8.71$, $p < .001$, $d = -1.23$, 95%CI $[-1.51, -0.95]$. Therefore, both manipulations were successful.

4.2.2. Effects of recommendation agent

We conducted a 2 (recommendation agent: algorithm, expert) $\times$ 2 (product type: search, experience) mixed ANOVA for mechanistic meta-dehumanization. The main effect of recommendation agent was significant, $F(1, 197) = 33.15$, $p < .001$, $\eta^2 = 0.14$, 95%CI $[0.08, 0.22]$, with algorithm recommendations ($M = 3.98$, $SD = 1.49$) eliciting more mechanistic meta-dehumanization compared with expert recommendations ($M = 3.17$, $SD = 1.37$). However, neither the main effect of

product type, $F(1, 197) = 1.79$, $p = .182$, $\eta^2 = 0.01$, 95%CI $[0.00, 0.04]$, nor the two-way interaction, $F(1, 197) = 0.05$, $p = .825$, $\eta^2 < 0.001$, 95%CI $[0.00, 0.01]$, was significant.

A similar ANOVA on intention to use revealed a null main effect of recommendation agent, $F(1, 197) = 2.04$, $p = .155$, $\eta^2 = 0.01$, 95%CI $[0.00, 0.05]$, a null main effect of product type, $F(1, 197) = 0.01$, $p = .908$, $\eta^2 < 0.001$, and no two-way interaction, $F(1, 197) = 1.53$, $p = .218$, $\eta^2 = 0.01$, 95%CI $[0.00, 0.04]$.

4.2.3. Mediating role of mechanistic meta-dehumanization

To examine the mediating role of mechanistic meta-dehumanization as a function of product type, we used Mplus 7.4 to test a multilevel moderated mediation model (Fig. 3). The model was singular, given that the random effect had a (nearly) zero-variance intercept; therefore, we used Bayesian estimation methods (Gelman & Hill, 2006).

The results show that algorithm (versus expert) recommendations increased mechanistic meta-dehumanization ($B = 0.81$, 95%CI $[0.52, 1.09]$, posterior $SD = 0.14$), which, in turn, predicted less intention to use ($B = -0.70$, 95%CI $[-0.81, -0.59]$, posterior $SD = 0.06$). When controlling for mechanistic meta-dehumanization and product type, algorithm (versus expert) recommendations elicited greater intention to use ($B = 0.36$, 95%CI $[0.13, 0.59]$, posterior $SD = 0.12$). Moreover, the two-way interactions on mechanistic meta-dehumanization ($B = 0.06$, 95%CI $[-0.50, 0.63]$, posterior $SD = 0.29$) and intention to use ($B = 0.39$, 95%CI $[-0.03, 0.82]$, posterior $SD = 0.22$) were not significant.

Study 3 provided further support for our hypothesis. Moreover, Study 3 suggested that consumers continue to display algorithm aversion, irrespective of the type of product being recommended. These findings extend the generalizability of our hypothesis.

5. Study 4

Study 4 aimed to provide a potential buffer against algorithm aversion based on the mediating role of meta-dehumanization. Perceptions that algorithms optimize interests aligned with consumers have been found to bolster people's warmth judgments about AI and willingness to cooperate with it (Khadpe et al., 2020; McKee et al., 2023). These favorable responses reflect perceptions of shared humanness with other agents (Haslam, 2022), constituting the springboard for the belief that one is humanized (a counterpart of being dehumanized). Therefore, we posited that when an algorithm's goal is framed as serving consumers (versus websites), it may buffer algorithm aversion caused by meta-dehumanization. In Study 4, we varied the frame (consumer- or website-serving) of algorithm recommendations. We predicted that

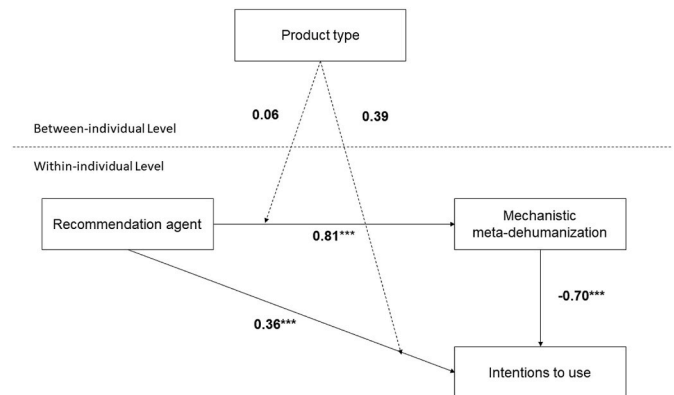


Fig. 3. Product type moderated the effects of recommendation agent on mechanistic meta-dehumanization and intention to use in Study 3.

Note. Recommendation agent manipulation was coded as follows: algorithm recommendation condition = 1, expert recommendation condition = 0. Product type manipulation was coded as follows: search product condition = 1; experience product condition = 0. Coefficients are unstandardized. *** $p < .001$.

framing algorithms as serving consumers (versus websites) would mitigate the indirect effect of recommendation agent on intention to use via mechanistic meta-dehumanization.

5.1. Method

5.1.1. Participants

Study 4 employed a 2 (recommendation agent: algorithm vs. expert) $\times$ 2 (algorithm frame: consumer-vs. website-serving) mixed-factorial design with recommendation agent as the within-subject factor and algorithm frame as the between-subject factor. Using the effect size ($d = 0.41$) obtained in Study 3 and $\alpha = .05$ (two-tailed), a sample size of 49 would be sufficient to achieve 80% power to detect the effect of recommendation agent on meta-dehumanization in a two-condition within-subject experiment. To explore whether the effect of recommendation agent varies across algorithm frames, we recruited 260 Chinese participants via Credamo and paid 2 CNY (0.29 USD). We excluded 45 participants following the preregistered criteria (i.e., being an outlier or failing attention check items). The final sample included 215 participants (148 women and 67 men, $M_{age} = 31.84$, $SD_{age} = 10.82$). Participants were randomly assigned to either the consumer- ($n = 99$) or website-serving ($n = 116$) frame condition.

5.1.2. Procedure and materials

5.1.2.1. Recommendation agent manipulation. After providing informed consent, the participants read materials similar to those in Study 3, with slight modifications. To introduce the algorithm frame manipulation, we changed the order of the descriptions of the two recommendation systems. Participants completed the same manipulation check questions as those in Study 3.

5.1.2.2. Algorithm frame manipulation. Next, participants were randomly assigned to one of two conditions. In the *consumer-serving frame condition*, participants read that algorithm recommendation systems were trained to satisfy consumers' needs and interests (e.g., helping consumers find their favorite products and reducing disturbances from uninteresting information). In contrast, in the *website-serving frame condition*, participants read that algorithm recommendation systems are trained to increase e-commerce websites' profits and popularity (e.g., recommending sponsored products and selling expensive products to regular consumers). As a manipulation check, participants answered three questions (e.g., "According to the material, to what extent do you think the goals of algorithm recommendation systems are aligned with consumers or websites"; 1 = *aligned with website goals*, 4 = *equal*, 7 = *aligned with consumer goals*). We averaged the three ratings to create the algorithm frame index ($\alpha = .92$). Higher ratings indicated that the algorithm was framed as serving consumers more than websites.

5.1.2.3. Meta-dehumanization. After that, participants completed the same scale measuring mechanistic meta-dehumanization as used in Study 3 (algorithm: $\alpha = .94$; expert: $\alpha = .94$). The presentation order of the two scales was randomized.

5.1.2.4. Intention to use. Finally, for each recommendation agent, participants reported their intention to use, as in Study 3. We averaged relevant items to create respective indices for intention to use algorithm ($\alpha = .94$) and expert ($\alpha = .92$) recommendations.

5.2. Results and discussion

5.2.1. Manipulation checks

Participants believed that algorithm recommendations ($M = 1.28$, $SD = 0.50$) worked less based on experience, compared with expert recommendations ($M = 6.64$, $SD = 0.59$; $t(214) = -79.42$, $p < .001$, $d =$

-5.42 , 95%CI $[-5.55, -5.28]$). Furthermore, participants in the consumer-serving goal condition ($M = 5.59$, $SD = 1.33$) perceived the algorithm recommendation goals as being more aligned with consumers (versus websites) than their counterparts in the website-serving goal condition ($M = 2.02$, $SD = 1.08$; $t(187.63) = 21.35$, $p < .001$, $d = 2.95$, 95%CI $[2.67, 3.22]$). Therefore, both manipulations were successful.

5.2.2. Effects of recommendation agent

We conducted a mixed ANOVA for mechanistic meta-dehumanization. The main effect of algorithm frame was significant, $F(1, 213) = 8.65$, $p = .004$, $\eta^2 = 0.04$, 95%CI $[0.01, 0.09]$, with the website-serving frame ($M = 4.06$, $SD = 1.23$) triggering more mechanistic meta-dehumanization than the consumer-serving frame ($M = 3.56$, $SD = 1.31$). The main effect of recommendation agent was also significant, $F(1, 213) = 34.98$, $p < .001$, $\eta^2 = 0.14$, 95%CI $[0.08, 0.21]$, with algorithm recommendations ($M = 4.26$, $SD = 1.73$) eliciting more mechanistic meta-dehumanization than expert recommendations ($M = 3.40$, $SD = 1.61$). Moreover, the two-way interaction was significant, $F(1, 213) = 32.90$, $p < .001$, $\eta^2 = 0.13$, 95%CI $[0.07, 0.21]$. Further simple effects analyses (Fig. 4A) showed that, in the website-serving frame condition, participants reported more mechanistic dehumanization by algorithm recommendations ($M = 4.86$, $SD = 1.62$) than expert recommendations ($M = 3.27$, $SD = 1.58$), $t(213) = 8.58$, $p < .001$, $d = 0.80$, 95%CI $[0.62, 0.98]$. However, in the consumer-serving frame condition, participants reported similar levels of mechanistic meta-dehumanization by algorithm ($M = 3.57$, $SD = 1.60$) and expert recommendations ($M = 3.54$, $SD = 1.63$), $t(213) = 0.12$, $p = .903$, $d = 0.01$, 95%CI $[-0.19, 0.21]$.

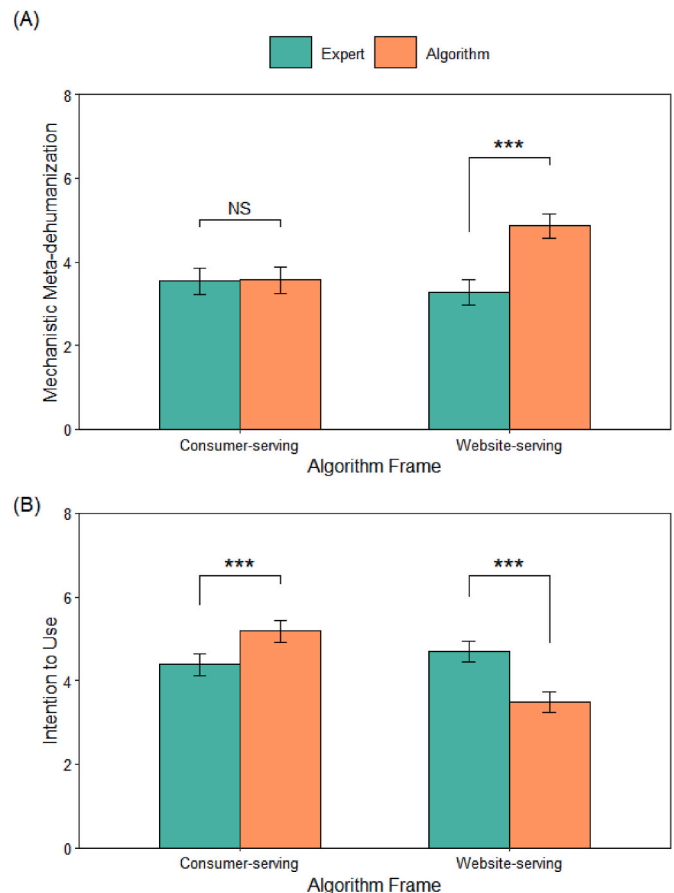


Fig. 4. Recommendation agent on (A) mechanistic meta-dehumanization and (B) intention to use under different algorithm frames in Study 4.

Note. Error bars depict 95% confidence intervals of means. NS = non-significant. *** $p < .001$.

We conducted a similar ANOVA for intention to use. The website-serving frame ($M = 4.08$, $SD = 0.97$) elicited less intention to use recommendation systems than the consumer-serving frame ($M = 4.78$, $SD = 0.91$), $F(1, 213) = 28.48$, $p < .001$, $\eta^2 = 0.12$, 95%CI [0.06, 0.19]. However, participants reported similar levels of intention to use algorithm ($M = 4.26$, $SD = 1.60$) and expert recommendations ($M = 4.55$, $SD = 1.38$), $F(1, 213) = 2.47$, $p = .117$, $\eta^2 = 0.01$, 95%CI [0.00, 0.05]. As expected, the two-way interaction was significant, $F(1, 213) = 56.43$, $p < .001$, $\eta^2 = 0.21$, 95%CI [0.14, 0.29]. Further simple effect analyses (Fig. 4B) showed that the website-serving frame was used, participants were less willing to use algorithms ($M = 3.48$, $SD = 1.51$) than experts ($M = 4.70$, $SD = 1.31$) for recommendations, $t(213) = -6.69$, $p < .001$, $d = -0.62$, 95%CI [-0.81, -0.44], indicating algorithm aversion. However, when using the consumer-serving frame, participants were more willing to use algorithm ($M = 5.18$, $SD = 1.15$) than expert ($M = 4.38$, $SD = 1.45$) recommendations, $t(213) = 4.04$, $p < .001$, $d = 0.41$, 95%CI [0.21, 0.61], indicating algorithm appreciation (Logg et al., 2019).

5.2.3. Mediating role of mechanistic meta-dehumanization

To examine the mediating role of mechanistic meta-dehumanization as a function of algorithm frame, we used Mplus 7.4 to test a multilevel moderated mediation model. The model was singular, given that the random effect had a (nearly) zero-variance intercept; therefore, we used Bayesian estimation methods (Gelman & Hill, 2006). The results (Fig. 5) show that algorithm (versus expert) recommendations increased mechanistic meta-dehumanization ($B = 0.86$, 95%CI [0.59, 1.14], posterior $SD = 0.14$), which, in turn, predicted less intention to use ($B = -0.66$, 95%CI [-0.76, -0.56], posterior $SD = 0.05$). When controlling for mechanistic meta-dehumanization and algorithm frame, algorithm (versus expert) recommendations elicited greater intention to use ($B = 0.28$, 95%CI [0.06, 0.50], posterior $SD = 0.11$).

Moreover, the two-way interactions on mechanistic meta-dehumanization ($B = -1.56$, 95%CI [-2.10, -1.01], posterior $SD = 0.28$) and intention to use ($B = 1.00$, 95%CI [0.57, 1.43], posterior $SD = 0.22$) were significant. Most importantly, the difference between the two types of algorithm frames in the indirect effect via mechanistic meta-dehumanization was significant, $B = 1.03$, 95%CI [0.65, 1.44], posterior $SD = 0.20$. Specifically, when framing the algorithm as website-serving, algorithm (versus expert) recommendations decreased intention to use via increased mechanistic meta-dehumanization, $B = -1.08$, 95%CI [-1.40, -0.80], posterior $SD = 0.15$. However, when framing the algorithm as consumer-serving, the indirect effect via meta-

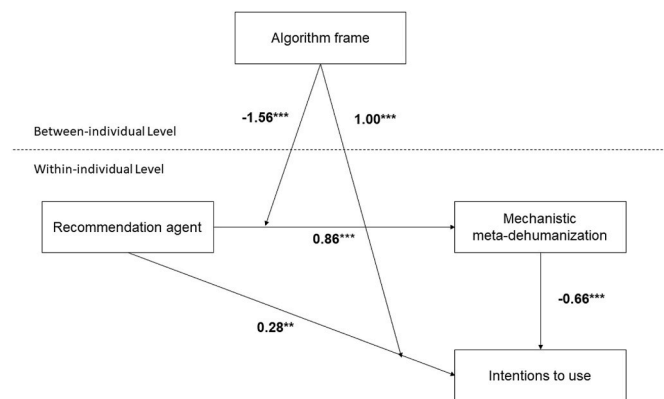


Fig. 5. Algorithm frame moderated the effects of recommendation agent on mechanistic meta-dehumanization and intention to use in Study 4.

Note. Recommendation agent manipulation was coded as follows: algorithm recommendation condition = 1, expert recommendation condition = 0. Algorithm frame manipulation was coded as follows: consumer-serving frame condition = 1, website-serving frame condition = 0. Coefficients are unstandardized. ** $p < .01$. *** $p < .001$.

dehumanization was non-significant, $B = -0.06$, 95%CI [-0.31, 0.20], posterior $SD = 0.13$.

Study 4 extended the hypothesis by identifying a potential intervening factor. Framing the algorithm as consumer-serving (versus website-serving) mitigated the algorithm aversion caused by mechanistic meta-dehumanization. The findings have practical implications for e-commerce businesses seeking to reduce consumer aversion to algorithms when opening the black box of algorithms by introducing their calculative nature. We return to this point in the General Discussion.

6. General discussion

The black box problem of algorithms pervades various contexts, including healthcare, education, and legislation (Bonezzi et al., 2022; Durán & Jongsma, 2021; Gillani et al., 2023), inducing algorithm aversion among users (Cadario et al., 2021; Yeomans et al., 2019). Although opening the black box is an intuitive solution for mitigating algorithm aversion, its effectiveness hinges on the approach employed (De Freitas et al., 2023; Koo et al., 2015; Larasati et al., 2020). Drawing from the notion of meta-dehumanization, we argued that algorithm aversion may persist when opening the black box by introducing algorithms' calculative nature.

Across four preregistered studies, we found that, compared to expertise-based human recommendations, calculation-based algorithm recommendations elicited more mechanistic meta-dehumanization, which in turn, predicted less intention to use. Notably, we clarified that the effect exists regardless of the type of algorithms (LLMs or conventional algorithms; Study 2) or recommended product (search or experience; Study 3). However, considering LLMs (versus conventional algorithms) as the recommendation agent (Study 2) and framing algorithms as consumer-serving (versus website-serving) helps mitigate algorithm aversion (Study 4).

Unlike previous research that has shown consumers to have an explicit resistance to algorithms in favor of humans (e.g., Dietvorst et al., 2015; Longoni et al., 2019), our studies did not yield significant total effects of recommendation agent on intention to use. In contrast, our results reveal that people are less inclined to use recommendations from algorithms (versus those from humans) because of increased meta-dehumanization. The lack of total effects, combined with the presence of indirect effects through meta-dehumanization, implies the co-existence of algorithm aversion and appreciation (Logg et al., 2019; Pezzo & Beckstead, 2020; Schaap et al., 2023; Williams et al., 2019). Notably, the mediating role of meta-dehumanization we observed provides empirical evidence for the anecdotal speculation that people find algorithms to be dehumanizing (Dawes, 1979; Grove & Meehl, 1996; Lee, 2018). Moreover, while previous literature has suggested that increasing transparency in algorithms could reduce aversion (Cadario et al., 2021; Shin, 2023; Yeomans et al., 2019), our findings indicate that revealing the calculative nature of algorithms may paradoxically heighten algorithm aversion.

6.1. Theoretical and practical implications

The present research contributes to ongoing discussions regarding the transparency of algorithms (Von Eschenbach, 2021). Prior research has demonstrated that explaining how algorithmic decisions are reached can help alleviate users' algorithm aversion (Cadario et al., 2021; Yeomans et al., 2019). However, our research found that, even when provided detailed explanations of an algorithm's calculative process, consumers still demonstrated aversion. This highlights the importance of the specific content of explanation-focused interventions (De Freitas et al., 2023). Offering explanations for the reasons behind algorithmic decisions, rather than focusing on the process, is more effective in building trust in algorithms (Koo et al., 2015). Therefore, elucidating the value of calculations-based algorithms could help mitigate algorithm aversion. For example, algorithms' perceived objectivity due to their

calculative nature is instrumental in addressing inequality; thus, explaining how algorithms ensure impartiality through calculations may lead people to prefer algorithm over human decision-making in unequal contexts (Bigman et al., 2021). Consequently, the content of explanations plays a nuanced role in opening the black box of algorithms.

The present research also provides novel insight into algorithm aversion. Unlike previous studies explaining algorithm aversion through algorithm- or user-related factors (e.g., Dietvorst & Bharti, 2020; Dietvorst et al., 2015), our findings reveal meta-dehumanization as a mechanism driving aversion to algorithm recommendations. These findings extend the anecdotal evidence that algorithms are dehumanizing (Grove & Meehl, 1996; Lee, 2018) and contribute to perspectives on human–algorithm interaction (Fogg, 2002). Moreover, because mechanistic meta-dehumanization is closely related to the calculative nature of algorithms, we believe that it acts as an inherent mechanism of algorithm aversion. Furthermore, mechanistic meta-dehumanization drives aversion to human-like algorithms (i.e., LLMs, Study 2). This indicates that users may feel mechanistically dehumanized and resist recommendation agents that are perceived to calculate their personal characteristics as data.

The present research encourages reconsideration of how people respond to algorithms. Prior research has found either algorithm appreciation (Logg et al., 2019; Williams et al., 2019) or aversion (Castelo et al., 2019; Longoni et al., 2019; Yeomans et al., 2019). Our findings present null total effects of recommendation agent on intention to use in Studies 1, 3 and 4 (but not in Study 2), suggesting the coexistence of algorithm appreciation and aversion (Jussupow et al., 2020). Further mediation analyses showed that the total effects could be divided into negative indirect effects (algorithm aversion) via mechanistic meta-dehumanization and positive direct effects (algorithm appreciation). As we demonstrated that mechanistic meta-dehumanization drives algorithm aversion, future research should explore why algorithm appreciation occurs. One possible reason might be that people experience a sense of agency in decision-making when algorithms function as advisors rather than delegates (Chugunova & Sele, 2020). Moreover, showing that algorithms can outperform humans and emphasizing their benefits and controllability may elicit algorithm appreciation (Pezzo & Beckstead, 2020; Schaap et al., 2023). Notably, in Study 2, wherein we specified a context of buying gifts, the results revealed both a negative total effect and negative indirect effect via mechanistic meta-dehumanization, without a positive direct effect. These findings suggest that when recommendations entail greater sociality, the perceptions of algorithms as lacking interpersonal warmth may be more pronounced in users' attitudes toward algorithms. Thus, future research should further examine this possibility.

The present research extends the literature on meta-dehumanization to the new and rapidly growing field of human–algorithm interaction. Previous research has highlighted the cognitive effects of algorithms on users' information processing (Shin et al., 2024) and self-perception (Longoni et al., 2019; Shin et al., 2022). As our research shows, meta-dehumanization represents another outcome of using algorithms at the meta-cognitive level. Algorithms' calculative working processes may lead users to perceive their humanity as devalued, resulting in various negative outcomes such as reduced self-esteem (Demoulin et al., 2021). This raises important concerns regarding the consequences of unveiling the inner workings of algorithms. Additionally, although most studies have focused on meta-dehumanization among humans (e.g., Bastian & Haslam, 2011; Kteily et al., 2016), the perpetrators of dehumanization may not always be humans. For example, disruptive technology may be dehumanizing, because it invades human autonomy and privacy to some extent (Oviatt, 2021, pp. 278–287; Zardiashvili & Fosch-Villaronga, 2020). We found that algorithms can be considered perpetrators of dehumanization because they calculate humanness.

On a practical level, the present research can provide inspiration for e-commerce businesses regarding how they can present their algorithm recommendations to consumers to reduce algorithm aversion.

Businesses should avoid referring to algorithms' calculative nature when explaining how they function, as the present research suggested that algorithm aversion persists under such explanations. They could instead introduce algorithms as being meticulously trained by engineers to provide recommendations based on accumulated experience. Another promising intervention to counter algorithm aversion is applying interactive AI technologies in recommendation agents. For example, researchers are developing systems that use prompts to combine conventional recommendation algorithms with LLMs (Gao et al., 2023), and these new systems are more likely than conventional algorithms to be adopted as they provide a more interactive and less technical approach.

Alternatively, highlighting goal alignment between algorithm recommendations and consumers may be effective in reducing algorithm aversion. For example, portraying algorithms as designed to enhance consumer satisfaction rather than solely focusing on marketing efficiency or profits may increase consumer acceptance (McKee et al., 2023). Interventions aimed toward developing a win-win mindset regarding consumer–algorithm relationships may also promote algorithm adoption. Businesses can educate consumers on how algorithm applications can not only help them find the product they want more efficiently but also enhance the overall effectiveness of e-commerce marketing.

6.2. Limitations and future directions

This research provides a novel perspective on the black box problem; however, it is not without limitations. First, the manipulation of recommendation agents could be more fine-grained. In the present research, we included calculation-based algorithms and experience-based experts; however, algorithms may incorporate human thinking in their functions, while human experts may rely on calculations to develop recommendations. To clarify whether the observed effects are rooted in the working method of algorithms (calculation versus expertise) or the recommendation agents (algorithms versus experts), we conducted a supplemental study (reported in the Supplementary Material). This study included three recommendation agents: calculation-based algorithms, experience-based experts, and calculation-based experts. The mechanistic meta-dehumanization scores in the calculation-based algorithm condition were similar to those in the calculation-based expert condition, but higher than those in the experience-based expert recommendation condition. These findings indicate that, for algorithm aversion, calculation-based explanations of working processes might be more critical than the recommendation agent. In addition, explanations of calculation-based work processes are typically more abstract than those of experience-based processes, potentially resulting in lower acceptance of algorithms compared to humans. Therefore, our effects may also be explained by participants' lack of understanding of the explanation/algorithm working process, instead of their aversion to the agent using calculative processes (Cadario et al., 2021).

Relatedly, we administered manipulation checks (e.g., asking participants to evaluate the recommendation agents' working processes) prior to measuring the dependent variables. This approach may raise concerns regarding whether it was the manipulation checks, rather than the manipulations themselves, that caused the observed effects. While our intention was to prompt participants to recall the materials they read, future studies could validate our results by either presenting manipulation checks after measuring the dependent variables or by not utilizing manipulation checks at all.

Second, the research highlights the potential drawbacks of explaining algorithm recommendation processes, but does not explore whether calculation-based recommendations (versus expertise-based human recommendations) could produce positive outcomes, such as improved perceptions of fairness that may ultimately reduce aversion. Third, we used limited sample sizes when testing our hypothesis in the context of product recommendations. Future research could recruit more participants and extend the generalizability of our findings to other contexts in

which human identity is less threatened and the evaluative criteria are more objective than subjective, such as for providing financial advice or scheduling events (Morewedge, 2022). Finally, our findings are mainly based on hypothetical scenarios and self-report measures, which limit the external validity. A direction for future research is to capture people's behavior in real-world shopping situations.

7. Conclusion

Algorithms function efficiently because of their high capacity to make powerful calculations. However, this advantage does not necessarily elicit appreciation among users, and may even result in aversion, especially when an algorithm's calculative nature is introduced by opening the black box. Understanding whether opening the black box of algorithms improves users' attitudes is crucial, because public reactions to algorithms largely determine whether society can benefit from these innovative technologies. Across four preregistered studies, we found that mechanistic meta-dehumanization leads users to avoid using algorithms. Our research contributes to the ongoing discourse surrounding algorithm transparency and aversion. Simply opening the black box of algorithms is insufficient; how transparency is achieved requires deeper scrutiny. By shifting the focus away from algorithms' purely calculative aspects, emphasizing the alignment of goals between businesses and consumers, and utilizing LLMs as recommendation agents, businesses can enhance consumers' algorithms acceptance. Taken together, the nuanced psychological processes underlying attitudes toward transparent algorithms warrant further theoretical and practical considerations.

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CRediT authorship contribution statement

Gewei Chen: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Jianning Dang:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Li Liu:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We made all data available at https://osf.io/2bfwx/?view_only=d992420c230448a186ada66a39547795.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chb.2024.108411>.

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